MCQ A · consolidation (after S6)

After S6.

For each question, only one option is correct. Justify your choice c
looking at the solutions.

1. In a basis, $(2;4;-2)$ and $(1;2;-1)$ are:

A. orthogonal

B. collinear

C. non-coplanar

D. equal

2. Three vectors $\vec{u}, \vec{v}, \vec{w}$, with $\vec{u}, \vec{v}$ non-collinear, are coplanar if:

A. $\vec{w} = \alpha\vec{u} + \beta\vec{v}$

B. $\vec{w} = 0$ only

C. their norms are equal

D. they have three coordinates

3. Two lines in space with non-collinear direction vectors are necessarily:

A. intersecting

B. non-coplanar

C. either intersecting or non-coplanar

D. parallel

4. In a basis, the expression of a vector is:

A. always impossible

B. unique when the three vectors form a basis

C. independent of the basis

D. only for orthonormal coordinate systems

5. An edge hidden in a perspective drawing must be:

A. erased

B. confused with a diagonal

C. shown with a different line style

D. declared not to exist

MCQ B · geometry review (after S17)

After S17.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. To show that $M \in (ABC)$, for A, B, C not collinear, we can:

A. compare AM to AB

B. write $\overrightarrow{AM} = s\overrightarrow{AB} + t\overrightarrow{AC}$

C. compute an area only

D. look at the figure

7. If two planes have the same direction, they are:

A. perpendicular

B. parallel or coincident

C. necessarily coincident

D. intersecting

8. In an orthonormal coordinate system, the midpoint of $A(0;1;2)$ and $B(2;3;4)$ is:

A. $(1;2;3)$

B. $(2;4;6)$

C. $(1;1;2)$

D. $(0;2;3)$

9. Two lines whose projections cross on the page:

A. intersect

B. are parallel

C. may be non-coplanar

D. coincide

10. For a triple to be a basis of space, it is enough to check:

A. non-coplanarity

B. orthogonality

C. equal norms

D. the initial point of the vectors

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Two non-parallel lines in space intersect.

2. The zero vector is collinear with every vector.

3. A perspective drawing can serve as a proof of coplanarity.

Review of prior work

I can choose a basis, compute a linear combination and distinguish relative positions.

To watch out for:
mistaking an appearance of the figure for a property of space; forgetting a coordinate equation.